

Customer Retention & RFM Analytics:

A Multi-Dimensional Framework for Strategic Customer Segmentation



1. Business Background

NovaMart is a fast-growing, multi-vendor e-commerce marketplace connecting thousands of independent third-party sellers with consumers across diverse product categories. Over its operational history, the platform has scaled rapidly—processing approximately **\$16 Million in gross revenue** across nearly **96,000 unique customers**.

During its initial growth phase, NovaMart prioritized **top-line customer acquisition** and gross merchandise value (GMV) expansion. However, as customer acquisition costs (CAC) across digital channels began to rise, leadership recognized that sustainable profitability could no longer rely solely on bringing new first-time buyers onto the platform.

NovaMart's underlying data ecosystem spans **9 relational operational tables**—capturing granular records across orders, order line items, payment installments, customer locations, product catalogs, seller profiles, and post-delivery customer reviews. Despite sitting on this rich transactional data, the business lacked a unified customer-level view to understand **post-purchase buyer behavior, repeat purchase velocity**, and the **operational factors** driving customers away after their first order.

2. Problem Statement

Despite generating roughly **\$16M in total revenue** and acquiring **~96,000 unique customers**, NovaMart faces a severe structural retention crisis: **an alarming ~97% customer churn rate**, meaning only ~3% of customers ever return to place a second order.

This core business challenge is compounded by three critical blind spots:

1. **The "Leaky Bucket" Acquisition Trap:** Marketing budgets are spent acquiring new users who treat NovaMart as a one-time transactional marketplace rather than a habitual shopping destination.
2. **Undifferentiated Marketing Spend:** Without behavioral segmentation, promotional campaigns treat a high-spending recent buyer the exact same way as a low-value buyer who has been inactive for over a year—resulting in wasted discount margins and low conversion rates.
3. **Hidden Operational Friction:** Customer retention is treated purely as a marketing problem, ignoring how post-purchase logistical bottlenecks—specifically **extended delivery timelines**—damage customer satisfaction and destroy repeat purchase intent before a second transaction can occur.

Core Analytical Objective:

Engineer an end-to-end data pipeline and interactive business intelligence solution (using **Excel**, **PostgreSQL**, and **Power BI / DAX**) to segment the ~96,000 customer base via an **RFM (Recency, Frequency, Monetary)** framework, diagnose the operational drivers behind the 97% churn rate, and deliver a targeted retention roadmap.

3. Executive Summary

This project establishes a centralized customer intelligence and retention framework for NovaMart by consolidating 9 raw transactional datasets in **PostgreSQL** and deploying a dynamic **Power BI Star Schema** powered by a custom **DAX RFM scoring engine**.

Key Analytical Findings

- **Severe Single-Order Concentration:** Out of ~96,000 unique customers (customer_unique_id), roughly **97% purchased only once**. Because transactional tables originally tracked order-specific IDs (customer_id), previous reporting masked the true repeat purchase rate.
- **Pareto Revenue Skew Across RFM Cohorts:** By scoring every customer on a 1-to-5 scale across Recency, Frequency, and Monetary value, the model revealed that a compact tier of **Champions** and **Loyal Customers** drives a disproportionate share of repeat revenue and exhibits significantly higher Average Order Values (AOV).
- **The 21-Day Delivery Satisfaction Cliff:** Cross-analyzing operational logistics against customer review ratings revealed a direct link between fulfillment speed and churn. When delivery times are binned into 7-day intervals (0–7 days, 8–14 days, 15–21 days, and 22+ days), customer satisfaction remains stable for the first two weeks but **plummets sharply once delivery exceeds 21 days**—identifying slow logistics as a primary catalyst for one-time buyer churn.

Business Impact & Expected Outcome

By transitioning from blanket promotions to **RFM-driven lifecycle marketing** and enforcing a **maximum 14-to-21-day delivery SLA**, NovaMart can systematically convert single-order buyers into repeat customers, recover high-value "At-Risk" accounts, and unlock significant incremental revenue without increasing acquisition spend.

4. Key Metrics Framework

To ensure that retention initiatives drive profitable growth without breaking unit economics or seller operations, the project defines a hierarchy of **North Star**, **Driver**, and **Guardrail** metrics.

Metric Category	Metric Name	Strategic Role in Project
North Star Metric (NSM)	Repeat Purchase Rate & Active High-Value Cohort Share	Measures whether NovaMart is successfully shifting from one-time transactions to recurring customer lifetime value (CLV).
Primary Driver KPI	Total Revenue (Monetary)	Tracks top-line financial scale and total monetary contribution per segment.
Primary Driver KPI	Average Order Value (AOV)	Evaluates basket size efficiency and spending power across RFM segments.
Primary Driver KPI	Recency (Days Since Last Order)	Identifies buyer warmth and urgency for retention or win-back triggers.
Guardrail Metric	Customer Churn Rate	Ensures acquisition growth does not mask underlying retention decay.
Guardrail Metric	Delivery SLA Breach Rate (21+ Days)	Protects customer experience; prevents marketing from spending money to re-acquire users who churned due to bad shipping.
Guardrail Metric	Average Review Score (CSAT)	Monitors product and fulfillment quality across sellers and shipping intervals.

5. Steps Taken in Analysis(EDA)

Rather than jumping straight into dashboard visuals, the analysis followed a structured, four-stage data engineering and analytics workflow:

Step 1: Data Profiling & Grain Auditing (Microsoft Excel)

- Audited all 9 raw source datasets (orders, order_items, order_payments, order_reviews, customers, products, sellers, geolocation, and category_translation).
- Identified a critical schema nuance: customer_id is generated uniquely for *every single order*, whereas customer_unique_id represents the *actual individual buyer*. Using customer_id would falsely show 0% repeat purchases; mapping everything to customer_unique_id was established as the foundation of the analysis.
- Checked for missing timestamps (e.g., unshipped orders missing order_delivered_customer_date), duplicate payment rows caused by multi-installment credit card transactions, and extreme monetary outliers.

Step 2: Relational Data Modeling & Master Table Engineering (PostgreSQL)

- Loaded the raw relational tables into **PostgreSQL** to handle complex aggregations prior to BI ingestion.
- Constructed Common Table Expressions (CTEs) and multi-table SQL JOIN pipelines to aggregate payment totals at the order level first—preventing fan-out duplicate revenue traps when joining order_items with order_payments.
- Engineered a clean **Master Analytical Fact Table** combining customer identifiers, order timestamps, total payment value, item counts, product categories, delivery duration (in days), and review scores.

Step 3: Star Schema Design & DAX Engineering (Power BI)

- Imported the curated PostgreSQL views into **Power BI** and structured a high-performance **Star Schema** connecting fact transactions to dimension tables (Dim_Customer, Dim_Product, Dim_Date, Dim_Location).
- Authored foundational DAX measures (Total Revenue, Unique Orders, Unique Customers, AOV, Avg Delivery Days, Avg Review Score).
- Built a customer-level summary table and DAX scoring engine to assign **1–5 rankings** for Recency, Frequency, and Monetary metrics, then mapped combined scores into human-readable customer segments using SWITCH(TRUE(), ...) conditional logic.

Step 4: Operational Bottleneck & Cohort Cross-Analysis

- Analyzed revenue distribution, order frequency, and AOV across each RFM segment.

- Grouped order delivery times into **7-day bins** (0–7 Days, 8–14 Days, 15–21 Days, 22+ Days) and correlated them against average customer review scores and repeat purchase rates to uncover *why* customers were churning.
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6. Hypothesis Formulation & Testing

To move beyond descriptive reporting into diagnostic analytics, three core business hypotheses were formulated and tested during the investigation:

Hypothesis 1: The Retention vs. Acquisition Asymmetry

- **Hypothesis Statement:** *NovaMart's ~\$16M revenue is heavily reliant on one-time transactions, and customer value is concentrated within a small fraction of repeat buyers.*
- **Test Method:** Aggregated distinct order counts per customer_unique_id in PostgreSQL and analyzed cohort revenue contribution in Power BI.
- **Result (Validated):** Confirmed that **~97% of customers placed only 1 order**. The remaining ~3% of repeat buyers generated substantially higher lifetime value and higher AOV—proving that even a 1% to 2% shift in repeat conversion would yield massive revenue gains without extra ad spend.

Hypothesis 2: Logistical Delays as a Hidden Churn Catalyst

- **Hypothesis Statement:** *Customers who experience extended shipping delays assign significantly lower review scores and are far less likely to return for a second purchase.*
- **Test Method:** Calculated exact delivery lead time (Delivered Date - Purchase Date), bucketed orders into 7-day intervals (0–7, 8–14, 15–21, 22+ days), and plotted Average Review Score and Order Volume across bins.
- **Result (Validated):** Customers receiving orders within 0–14 days maintained high satisfaction ratings (above 4.0 stars). However, **once delivery duration crossed the 21-day threshold, average review scores collapsed sharply**. Poor logistics directly poisoned first-time buyer trust, explaining why so many customers never returned.

Hypothesis 3: Spend & Payment Behavior Across Segments

- **Hypothesis Statement:** *High-monetary customers exhibit distinct basket sizes (AOV) even when their purchase frequency is low, meaning some single-order buyers are high-value "Promising / Potential Loyalists" rather than low-value churners.*
- **Test Method:** Evaluated AOV and Monetary quintiles independently of Frequency within the RFM matrix.
- **Result (Validated):** Identified a sizable segment of recent, high-AOV single-order buyers who should be targeted with high-touch second-purchase onboarding rather than generic clearance discounts.

7. Why We Chose RFM & How It Was Calculated

How I Reached the Conclusion to Track RFM

During exploratory analysis, a major strategic problem emerged: **if 97% of customers only buy once, how do we prioritize who to target first?**

- If we only looked at **Frequency**, 97% of our ~96,000 customers would look identical (1 order).
- By introducing **Recency**, we immediately separated customers who bought *last week* (still warm, high probability of converting to a second order) from those who bought *a year ago* (cold, already churned).
- By introducing **Monetary**, we separated high-ticket buyers (worth investing margin and personalized offers into) from low-spend bargain hunters.

Combining **Recency, Frequency, and Monetary** gave NovaMart a three-dimensional behavioral lens to turn a flat list of 96,000 customers into actionable marketing tiers.

How RFM Was Calculated (SQL & DAX Implementation)

1. Establishing the Base Metrics per Unique Customer

For every customer_unique_id, three foundational values were calculated against the dataset's maximum reference date (Max_Dataset_Date):

- **Recency (\$R\$):** Days elapsed between the customer's most recent order_purchase_timestamp and the dataset's maximum date. (*Lower days = Higher score*).
- **Frequency (\$F\$):** Total count of distinct order_id values associated with the customer_unique_id. (*Higher count = Higher score*).
- **Monetary (\$M\$):** Total sum of payment_value spent by the customer_unique_id. (*Higher spend = Higher score*).

```
1  -- PostgreSQL Customer-Level RFM Base Aggregation
2  WITH reference_date AS (
3      SELECT MAX(order_purchase_timestamp)::DATE AS max_date
4      FROM orders
5      WHERE order_status = 'delivered'
6  )
7  SELECT
8      c.customer_unique_id,
9      (SELECT max_date FROM reference_date) - MAX(o.order_purchase_timestamp)::DATE AS recency_days,
10     COUNT(DISTINCT o.order_id) AS frequency_orders,
11     ROUND(SUM(p.payment_value)::NUMERIC, 2) AS monetary_value
12 FROM customers c
13 JOIN orders o ON c.customer_id = o.customer_id
14 JOIN order_payments p ON o.order_id = p.order_id
15 WHERE o.order_status = 'delivered'
16 GROUP BY c.customer_unique_id;
```

2. The 1–5 RFM Scoring Engine in Power BI (DAX)

In Power BI, each customer was assigned a score from **1 to 5** for Recency, Frequency, and Monetary using percentile/threshold logic:

- **Recency Score (1–5):** Customers in the most recent 20% of purchase dates receive a **5**; the oldest 20% receive a **1**.
- **Frequency Score (1–5):** Because 97% of users have 1 order, standard quintiles clump together. Therefore, custom conditional thresholds were applied (e.g., 1 order = base score, 2 orders = mid-tier score, 3+ orders = top score of 5) or combined with Recency/Monetary ranks.
- **Monetary Score (1–5):** Customers in the top 20% of total spend receive a **5**; the lowest 20% receive a **1**.

```
Customer_Segment = VAR R = Customer_RFM[R_Score]
                    VAR F = Customer_RFM[F_Score]
                    RETURN
                    SWITCH(TRUE(),
                        R >= 4 && F >= 4, "Champions",
                        R >= 3 && F >= 3, "Loyal Customers",
                        R >= 4 && F <= 2, "Recent & Promising",
                        R <= 2 && F >= 3, "At Risk",
                        R <= 2 && F <= 2, "Lost/Hybernating",
                        "Average")
```

8. Strategic Recommendations & Action Plan

Based on the RFM segmentation and the 21-day logistics bottleneck discovery, NovaMart should execute a two-pillar strategy combining **Targeted Lifecycle Marketing** and **Supply Chain SLA Optimization**.

Pillar A: Segment-Specific Retention Campaigns

1. Break the "One-and-Done" Wall (New / Promising & Potential Loyalists):

- *The Problem:* 97% of buyers never place a second order.
- *Action:* Trigger an automated **Post-Delivery Next-Order Bounce-Back** campaign exactly **7 days after successful delivery** (when satisfaction is highest). Offer a time-sensitive credit (e.g., "10% off your second order within 14 days") or complementary cross-category product recommendations.

2. Protect & Monetize Top Tiers (Champions & Loyal Customers):

- *The Problem:* Over-discounting to loyal buyers erodes margin unnecessarily.
- *Action:* Stop sending generic discount codes to Champions. Instead, enroll them in a **NovaMart VIP Loyalty Tier** featuring priority shipping, early access to seasonal sales, and referral rewards to turn them into brand advocates.

3. High-ROI Win-Back Interventions (At-Risk High Value):

- *The Problem:* High-AOV buyers who haven't purchased in 120+ days are slipping into permanent churn.
- *Action:* Deploy high-touch personalized win-back bundles or installment fee waivers specifically targeted at high-monetary dormant accounts, while avoiding costly paid retargeting on low-spend Hibernating / Lost users.

Pillar B: Operational & Logistics Optimization

1. Enforce a Strict 14-to-21-Day Maximum Delivery SLA:

- Since customer satisfaction collapses after **21 days**, audit seller-to-customer shipping lanes that routinely fall into the 22+ Days bucket. Prioritize local/regional sellers in search rankings when a buyer's geolocation is far from the default vendor.

2. Proactive Delay Mitigation ("Save the Customer Before the Bad Review"):

- Implement an automated webhook alert whenever an in-transit order crosses **Day 15** without delivery confirmation. Proactively send an apology message with a small wallet credit *before* the order crosses Day 21—softening dissatisfaction and preserving future retention.

9. Conclusion

The **NovaMart Customer Retention & RFM Analytics** project demonstrates that solving a **97% churn rate** requires looking beyond surface-level sales totals and connecting **customer behavior** with **operational execution**.

By re-architecting 9 fragmented datasets in **PostgreSQL** around the true unique customer grain (customer_unique_id), building a scalable **Star Schema in Power BI**, and deploying a **DAX-driven RFM segmentation engine**, this solution transforms ~\$16M of historical transaction records into an actionable decision system. Pairing targeted RFM lifecycle marketing with strict controls on the **21-day delivery bottleneck** equips NovaMart to systematically increase repeat purchase frequency, elevate customer lifetime value, and build durable, long-term profitability.
